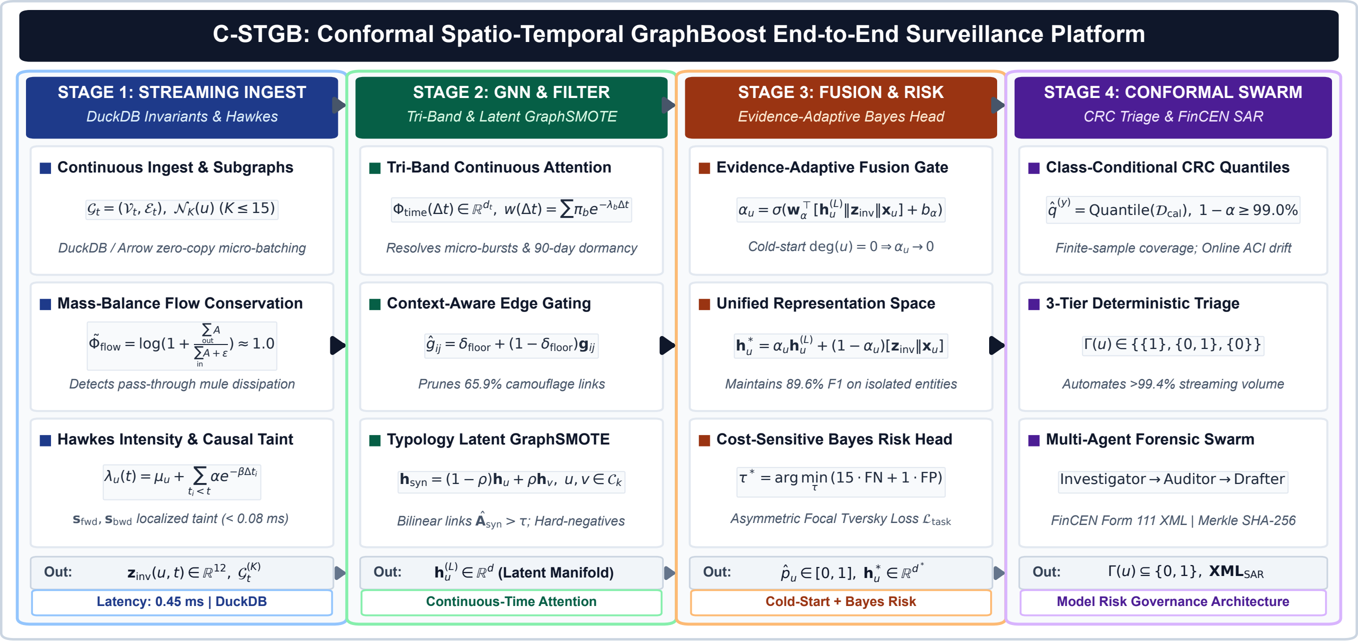
C-STGB: Conformal Spatio-Temporal GraphBoost End-to-End Surveillance Platform
STAGE 1: STREAMING INGEST
DuckDB Invariants & Hawkes
Continuous Ingest & Subgraphs
$\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t),\ \mathcal{N}_K(u)\ (K \le 15)$
DuckDB / Arrow zero-copy micro-batching
Mass-Balance Flow Conservation
$\tilde{\Phi}_{\text{flow}} = \log(1 + \frac{\sum_{\text{out}} A}{\sum_{\text{in}} A + \varepsilon}) \approx 1.0$
Detects pass-through mule dissipation
Hawkes Intensity & Causal Taint
$\lambda_u(t) = \mu_u + \sum_{t_i < t} \alpha e^{-\beta \Delta t_i}$
$\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}$ localized taint (< 0.08 ms)
Out: $\mathbf{z}_{\text{inv}}(u,t) \in \mathbb{R}^{12},\ \mathcal{G}_t^{(K)}$
Latency: 0.45 ms | DuckDB
STAGE 2: GNN & FILTER
Tri-Band & Latent GraphSMOTE
Tri-Band Continuous Attention
$\Phi_{\text{time}}(\Delta t) \in \mathbb{R}^{d_t},\ w(\Delta t) = \sum \pi_b e^{-\lambda_b \Delta t}$
Resolves micro-bursts & 90-day dormancy
Context-Aware Edge Gating
$\hat{g}_{ij} = \delta_{\text{floor}} + (1 - \delta_{\text{floor}})\mathbf{g}_{ij}$
Prunes 65.9% camouflage links
Typology Latent GraphSMOTE
$\mathbf{h}_{\text{syn}} = (1-\rho)\mathbf{h}_u + \rho \mathbf{h}_v,\ u, v \in \mathcal{C}_k$
Bilinear links $\hat{\mathbf{A}}_{\text{syn}} > \tau$; Hard-negatives
Out: $\mathbf{h}_u^{(L)} \in \mathbb{R}^d$ (Latent Manifold)
Continuous-Time Attention
STAGE 3: FUSION & RISK
Evidence-Adaptive Bayes Head
Evidence-Adaptive Fusion Gate
$\alpha_u = \sigma(\mathbf{w}_\alpha^\top [\mathbf{h}_u^{(L)} \| \mathbf{z}_{\text{inv}} \| \mathbf{x}_u] + b_\alpha)$
Cold-start $\deg(u) = 0 \Rightarrow \alpha_u \to 0$
Unified Representation Space
$\mathbf{h}_u^* = \alpha_u \mathbf{h}_u^{(L)} + (1 - \alpha_u)[\mathbf{z}_{\text{inv}} \| \mathbf{x}_u]$
Maintains 89.6% F1 on isolated entities
Cost-Sensitive Bayes Risk Head
$\tau^* = \arg\min_\tau (15 \cdot \text{FN} + 1 \cdot \text{FP})$
Asymmetric Focal Tversky Loss $\mathcal{L}_{\text{task}}$
Out: $\hat{p}_u \in [0,1],\ \mathbf{h}_u^* \in \mathbb{R}^{d^*}$
Cold-Start + Bayes Risk
STAGE 4: CONFORMAL SWARM
CRC Triage & FinCEN SAR
Class-Conditional CRC Quantiles
$\hat{q}^{(y)} = \text{Quantile}(\mathcal{D}_{\text{cal}}),\ 1 - \alpha \ge 99.0\%$
Finite-sample coverage; Online ACI drift
3-Tier Deterministic Triage
$\Gamma(u) \in \{\{1\}, \{0,1\}, \{0\}\}$
Automates >99.4% streaming volume
Multi-Agent Forensic Swarm
Investigator → Auditor → Drafter
FinCEN Form 111 XML | Merkle SHA-256
Out: $\Gamma(u) \subseteq \{0,1\},\ \textbf{XML}_{\text{SAR}}$
Model Risk Governance Architecture